

Predicting Buy and Sell Signals for Stocks using Bollinger Bands and MACD with the Help of Machine Learning

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Abstract— Investors usually take the decisions of buying and selling the stocks by evaluating a company's performance. These stock prices can be increased or decreased. Technical traders use many indicators to develop trading strategies. These tools can also help ordinary investors. With a little background knowledge, an ordinary person can also buy and sell signals at right time with huge profits and low losses. This is accomplished by combining two dominant technical indicators Moving Average Convergence Divergence and Bollinger bands indicators, to create strategies, and then forecasting the trend using the ARIMA Time Series model. This model is mainly useful for short term trading. This model is provided with 98% accuracy in predicting the trend.

Keywords— *technical indicator, trend, volatility, moving average, squeeze, bollinger bands, moving average convergence and divergence, Autoregressive Integrated Moving Average model*

Introduction

It is a mathematical pattern generated from historical data, such as the value, volume, and open interest of a security that is used by technical traders or investors to anticipate future price movements and make trading choices on a security. It derives a sequence of data points using historical price, volume, and open interest data through the application of a mathematical formula. By analysing historical information, technical analyst use technical indicators to predict future price developments [1].

The existing system called the Fundamental analysis strategy is primarily concerned with measuring a security's intrinsic value by examining related economic and financial factors. Stocks that are likely to a positive price increase are filtered using P/E ratios. It relies on the assumption that successful businesses will remain advantageous in the long run due to the market's fulfilling nature [2][3].

The proposed system called the Technical analysis strategy which forecasts future trend using time-series analysis on historical trends. Moving average convergence divergence, Bollinger Bands are some of the many

indicators used to study and analyse how the market is moving [4][5].

Investors need to have a clear idea about the entry and exit points for buying and selling the stocks. An incorrect entry or exit points can lead to huge loss and may miss large profits. The paper mainly focuses on generate accurate buy and sell signals for short term trading strategies. The existing system uses only a single indicator but a single indicator cannot predict accurate results. Instead two different powerful technical indicators are combined along with the forecasted trend to increase the accuracy of the proposed model.

Technicians must have a solid grasp of the technical indicators they employ in order to recognize misleading signals when they occur. As a checking method, many specialists choose to utilize a combination of technical indications [6] [7]. Because trading on erroneous signals may be quite costly, transactions are only made when a group of technical indicators agree on a predicted price change [8].

Bollinger bands and Moving Average Convergence Divergence indicators are plotted using Python as the fundamental programming language and will utilize Plotly of python which will assist us with plotting the graphs of the proposed methodology. Forecasting trend using ARIMA time series model is plotted by utilizing matplotlib library of python [9].

I. FINANCIAL BACKGROUND KNOWLEDGE

A. Moving Average Convergence Divergence Indicator (MACD)

MACD was developed in 1979 by Gerald Appel, is one of the most widely used indicators in technical analysis. Its design necessitates the use of three exponential moving averages (EMA), which are lagging indicators, to determine if a trend is continuing or reversing. By subtracting the slow moving average from the fast moving average, these lagging indicators are transformed into a momentum oscillator [10][11]. The term "fast" refers to a shorter moving average,

while "slow" refers to a longer moving average. The timeframes for generating the first indication in a typical configuration are 12 and 26 days:

$$MACD = EMA1 (12\text{-day closing prices}) - EMA2 (26\text{-day closing prices})$$

The second indication, termed signal line, smooths the first indicator using EMA once more. The signal line's typical day setting is 9:

$$Signal = EMA (9\text{-day MMACD})$$

When the more volatile indicator (MACD) crosses the less volatile one (Signal) from below, a buy signal is created. When the MACD indicator crosses the Signal indicator, a sell signal is generated.

B. Bollinger bands

John Bollinger created the Bollinger Bands. The Bollinger Bands are made up of three lines (Trading Geek, 2014). The middle band is a 20-period simple moving average of the usual price (TP). F standard deviations (usually 2) above and below the central band define the upper and lower bands. When the price volatility is higher or lower, the bands broaden and narrow, accordingly. When the price is approaching the top or lower band, a trend reversal is possible. The middle band serves as a degree of support or resistance [12]. Price objectives can alternatively be understood as the lower and upper bands. The top band becomes the price objective when the price bounces off the lower band and crosses the middle band.

$BOLU=MA(TP,n)+m*\sigma[TP,n]$
$BOLD=MA(TP,n)-m*\sigma[TP,n]$
$TP \text{ (typical price)}=(High+Low+Close)\div 3$
Where,
$BOLU=Upper \text{ Bollinger Band}$
$BOLD=Lower \text{ Bollinger Band}$
$MA=Moving \text{ average}$
$TP =typical \text{ price}$
$n=Number \text{ of days in smoothing period}$
$m=Number \text{ of standard deviations}$
$\sigma[TP,n]=Standard \text{ Deviation over last } n \text{ periods of } TP$

II. INPUT DATASETS

The model proposed in this paper is applied on four Indian stock datasets such as INFOSYS, AIRTEL, TCS stock data sets are used with technical indicators for generating strategies and RELIANCE stock dataset which is used for forecasting trend using ARIMA model [13]. The timeline of first three datasets is from January 2019 to December 2020 and for last dataset is from 1996 to 2020. These stock datasets consists of some set of attributes such as Date, Open, Close, Low, High, Volume and so on [14][15]. For example, the last five records of Infosys dataset is shown below.

	Date	Open	High	Low	Close	Adj Close	Volume
2018-07-02	2018-07-02	657.000000	670.000000	657.000000	667.349976	617.566162	4830144.0
2018-07-03	2018-07-03	665.950012	678.950012	665.875000	676.875000	626.380554	8127452.0
2018-07-04	2018-07-04	676.025024	676.325012	669.424988	672.599976	622.424438	5025722.0
2018-07-05	2018-07-05	673.549988	673.625000	634.049988	642.000000	594.107178	17019902.0
2018-07-06	2018-07-06	645.125000	650.924988	634.500000	642.125000	594.222839	11215356.0

Figure 1. First five records of INFOSYS stock dataset

A. Key Attributes

- Open* means the price at which a stock started trading when the opening bell rings.
- Close* refers to a price of individual stock when the stock exchange closed shop for the day.
- High* refers to the maximum price at which stock traded during a period.
- Low* refers to the minimum price at which stock traded during a period.
- Volume* is the total amount of trading activity.

III. METHODOLOGY

The objective of the paper is to propose a methodology that could generate whether there will be a buy signal or a sell signal in the next coming time period. This has been implemented effectively using technical analysis and time series model with pandas, matplotlib, plotly, numpy, statsmodels and pmdarima packages of open source programming language Python. The process of the proposed method is shown in Figure 2.

A. Pre-processing

It is the first stage in the data pre-processing process that we will discuss. As part of data pre-processing, it is important to correctly detect and manage missing values; otherwise, you run the risk of drawing incorrect and inaccurate conclusions and inferences from the data. You can handle the missing data in two ways: (1) Deleting a particular row (2) Replace it with zero.

B. Calculate BB and MACD attributes

- Bollinger band attributes

In order to visualize a Bollinger band, we first need to calculate its key attributes. First, calculate the rolling mean, rolling standard deviation and rolling volume by taking window size as 20 for Close and Volume columns in dataset. Second, use the above attributes to calculate Bollinger high

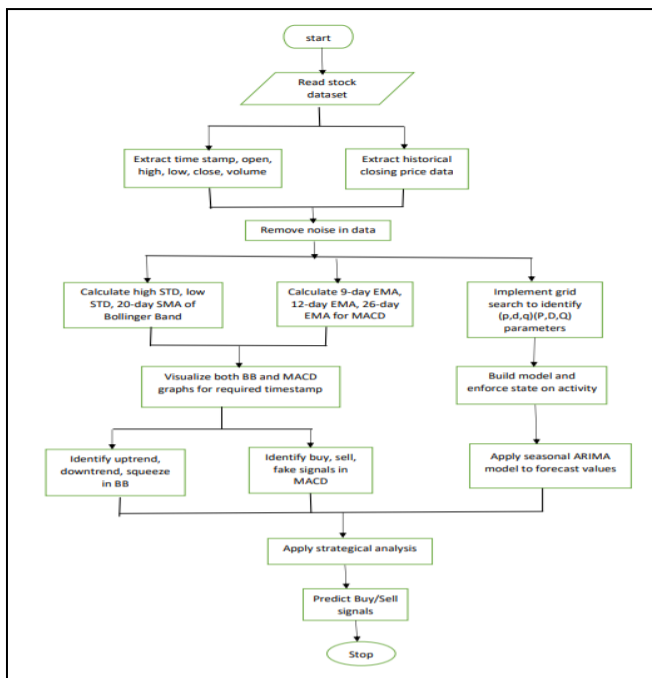


Figure 2. Process diagram of the proposed method

(upper band) and Bollinger low (lower band) attributes by setting number of standard deviations as 2. Third, take the mean of high, low and close values to calculate the moving average (represents the middle band of BB). Four, using the values of Bollinger high, Bollinger low, Open and Close attributes calculate the Bollinger signal.

$$\begin{aligned} \text{Rolling mean} &= \text{mean}(\text{rolling window}_{20}(\text{Close})) \\ \text{Rolling std} &= \text{std}(\text{rolling window}_{20}(\text{Close})) \\ \text{Rolling vol} &= \text{vol}(\text{rolling window}_{20}(\text{Close})) \end{aligned}$$

create two new DataFrame columns to hold values of upper and lower Bollinger bands:

$$\text{Bollinger High} = \text{Rolling mean} + (\text{Rolling std} * 2)$$

$$\text{Bollinger Low} = \text{Rolling mean} - (\text{Rolling std} * 2)$$

Calculate a new data frame attribute to hold the middle band:

$$TP = (\text{High} + \text{Low} + \text{Close}) / 3$$

$$MA = \text{mean}(\text{rolling window}_{20}(TP))$$

Rolling Mean	Bollinger High	Bollinger Low	Rolling Vol	TP	MA_TP	STD_TP	Bollinger SIGNAL
1144.735010	1264.133289	1025.336731	9900840.75	1241.433350	1143.987500	58.165342	0
1154.220014	1275.599562	1032.840465	9804726.70	1246.583333	1153.040835	59.387180	0
1162.945014	1284.617045	1041.272982	9705305.25	1246.083374	1161.955837	59.297835	0
1172.607514	1291.368686	1053.846341	9716589.65	1251.066691	1171.185004	58.006638	0
1180.932514	1299.820515	1062.044512	9144132.65	1260.583333	1179.601670	58.162726	0

Figure 3. Attributes of Bollinger band indicator of INFOSYS dataset

b. Moving average convergence divergence attributes

To visualize the MACD indicator, we need three main attributes they are: macd line, signal line and histogram. To calculate the macd line, and histogram we need to have the mean of moving averages of 26-day period and the 12-day period of Close column values. For signal line we must

calculate the mean of moving average of 9-day period of Close column.

To calculate 26-day and 12-day moving average,

$$\begin{aligned} \text{EMA_LONG} &= \text{mean}(\text{rolling window}_{26}(\text{Close})) \\ \text{EMA_SHORT} &= \text{mean}(\text{rolling window}_{12}(\text{Close})) \end{aligned}$$

create three new DataFrame columns to hold values of macd line, signal line and histogram

$$\begin{aligned} \text{MACD} &= \text{EMA_SHORT} - \text{EMA_LONG} \\ \text{SIGNAL} &= \text{mean}(\text{rolling window}_9(\text{Close})) \\ \text{HIST} &= \text{MACD} - \text{SIGNAL} \end{aligned}$$

EMA_LONG	EMA_SHORT	MACD	SIGNAL	HIST
1149.855698	1189.923044	40.067346	29.992510	10.074836
1157.296020	1199.211814	41.915794	32.377167	9.538627
1163.925948	1206.533081	42.607133	34.423160	8.183973
1170.731437	1214.112615	43.381177	36.214764	7.166414
1177.377253	1221.241436	43.864182	37.744647	6.119535

Figure 4. Attributes of MACD band indicator of INFOSYS dataset

C. Visualize BB and MACD indicators

Data visualization is an important part of data analysis. Python is one of the most popular programming languages for data visualization. Tis language provides various packages for data visualization. In this paper we discuss about Plotly package for the visualization of BB and MACD indicators. Plotly's python graphing library makes interactive publication quality graphs. Generally, visualization of Financial charts we use Time series and candlestick charts.

Bollinger band indicator chart consists of Upper band, Lower band, Middle band (20-day MA) and the price movement. The three bands are plotted using Time series graph and the price movement is plotted using candle stick graph in a single chart. Moving Average Convergence Divergence indicator consists of three lines signal line, macd line, and a histogram. The macd line and the signal line is plotted using Time series graph and the histogram is

plotted using histogram graph in a single chart. For example, the below diagram represents the Bollinger band and MACD graphs for Infosys data from January 2019 to June 2019.



Figure 5. Data visualization of Bollinger band indicator

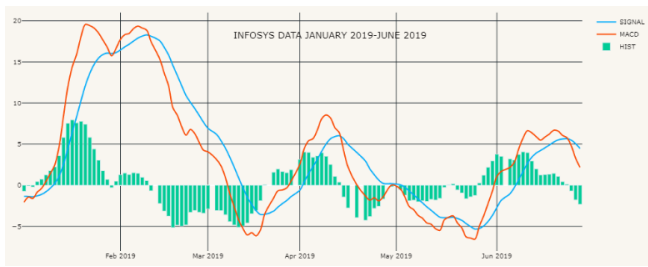


Figure 6. Data visualization of MACD indicator

D. Identification of Uptrend, Downtrend and Squeeze in Bollinger Band

This paper uses Bollinger band indicator to identify the uptrend, downtrend and the squeeze points from the price movement. An uptrend is used to generate a buy signal to make huge profits for short term trades. A downtrend generates a sell signal in order to avoid loss in trade. A squeeze point is a special condition generated in the Bollinger band where large breakouts occur.

- a. *Squeeze*: Squeeze is a pure reflection of volatility. When upper and lower bands are far apart then volatility is high and when upper and lower bands are close together then volatility is low. Squeeze point is triggered when the price movement is constant for a particular period of time during its low volatility. Break points such as overbought or oversold conditions can arise after a squeeze point. Hence forecasting stock price movements and the next buy or sell signal is an advantage of this concept.



Figure 7. Squeeze point of Bollinger band

- b. *Downtrend*: A downtrend is the movement of a stock's price towards a lower level than it was previously. It will exist as long as the stock market has a combination of lower highs and lower lows. The price will move along the bottom range in a severe decline, indicating that selling activity is still active. However, if the price falls to the lower range or moves along it, it indicates that the downtrend is losing steam.



Figure 8. Downtrend of Bollinger band

- c. *Uptrend*: When the price movement of a financial asset is entirely in the direction of upward movement, this is referred to be an uptrend. A rising trend has many peaks, each of which is higher than the preceding ones. For as long as the price maintains these higher lows and higher highs, the uptrend is deemed to be intact. Stock prices are considered overbought when they consistently touch the top band. Price fluctuates between the upper band and the 20-day moving average in a strong uptrend.



Figure 9. Uptrend of Bollinger band

E. Identification of Buy signal, Sell signal and Fake signal in MACD indicator

- a. *MACD Buy signal*: If the MACD crosses above its signal line after a brief rise higher inside a longer-term downtrend, traders would consider this to be a Buy confirmation, and the MACD crossed above its signal line.
- b. *MACD Sell signal*: Trading signals such as the MACD crossing below its signal line following a small rise upward inside a longer-term downtrend are seen as Sell confirmations by traders.

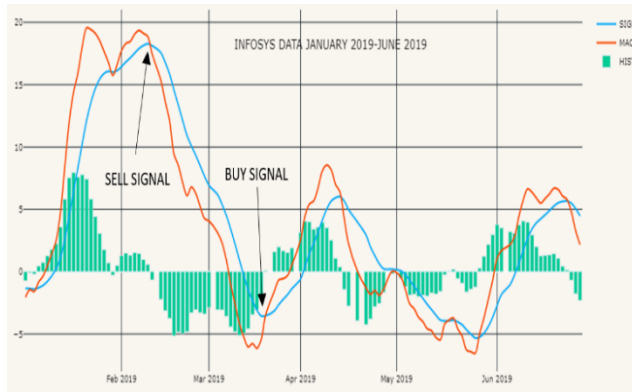


Figure 10. Buy and Sell signals of MACD indicator

- c. Fake signals: Divergence in the macd indicator generates fake signals. The primary difficulty with divergence is that it frequently suggests a reversal without really causing one to occur. It describes that when the direction of price action and entry-exit points are opposite to each other. In the below diagram, the macd line (red line) moves closer to the signal line (blue line) but no crossover occurs which indicates trend reversal which is a fake signal.

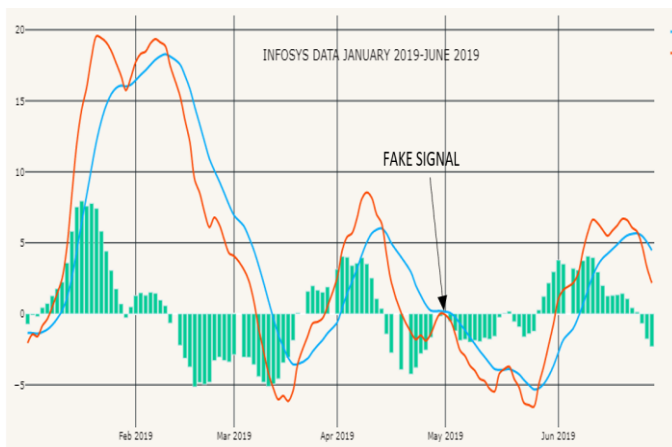


Figure 11. Fake signal of MACD indicator

F. Forecast trend using ARIMA model

ARIMA (Auto Regressive Integrated Moving Average) models are a sort of statistical model that may be used to analyse and forecast data that is presented in a time series format. ARIMA is a straightforward but effective technique for making expert time series forecasts. It does so by explicitly catering to a set of typical patterns in time series data, which is a basic but powerful approach.

ARIMA time series model uses Reliance stock dataset as input and produces forecasted trend as output. Below are the steps to forecast the trend.

- 1) Load dataset
- 2) ADF (Augmented Dickey-Fuller) Test for checking stationarity using null hypothesis and alternate hypothesis.
- 3) Since the data is not stationary Seasonal decompose with Additive model to make the model stationary.

- 4) Split the data into train and test set for creating ARIMA model.
- 5) Use auto ARIMA model to generate p, d, q parameters (generated (p, d, q) values for this model are (1, 1, 1)).
- 6) Create ARIMA model using optimal parameters.
- 7) Forecast the trend on test dataset with 95% confidence.
- 8) Check the accuracy metrics to judge accurate results.

IV. STRATEGICAL ANALYSIS

The model proposed in this paper uses MACD indicator to verify the accuracy of Bollinger band signal that evaluates a trading strategy. These strategies are taken from the analysis of three stock datasets (Infosys, Airtel, TCS) from various time periods.

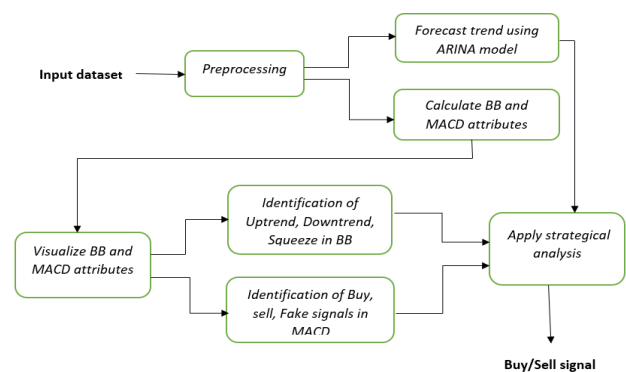


Figure 12. The overall framework for Generation of Buy and Sell signal

Since the analysis of Bollinger band indicator is based on forecast and prediction of future event and sometime may give us false signal it is therefore available to use other technical analysis to double check and confirm the results to enhance the accuracy of price trend buy and sell signal strategies. This model uses MACD indicator for the verification of Bollinger band indicator.

Strategy 1: Snap Back to Middle Line Strategy

Buy Signal: In a downward trend, if the price trend exits the lower Bollinger band and stop due to rapid decrease, it generates a buy signal and the possible reversal of the trend.



Figure 13: Snap Back to Middle line strategy for Buy signal

Sell Signal: In an upward trend, if the price trend exits the upper band due to rapid increase in price, it generates a sell signal and possible reversal of trend.

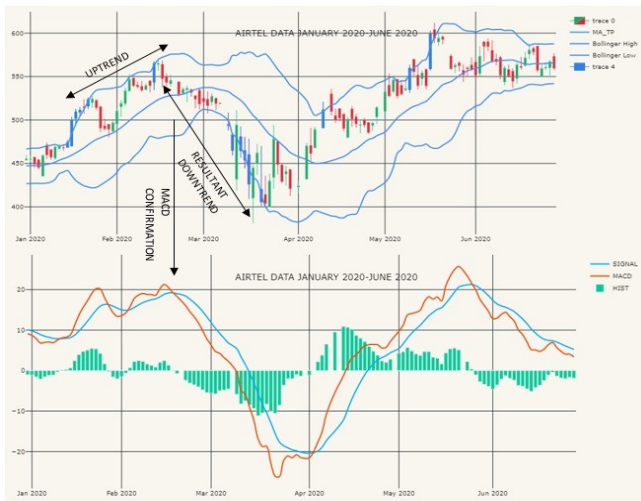


Figure 14: Snap Back to Middle Line Strategy for Sell signal

Strategy 2: Trend Reversal strategy

Buy Signal: In a downward trend, if the price trend exit the lower band with a bearish candle and closes below the lower band and then return back again and closes inside the lower band with bullish candle, then a buy signal is generated. MACD indicator is used for justification.



Figure 15: Trend Reversal Strategy for Buy Signal

Sell Signal: In an upward trend, if the price trend exit the upper band with a bullish candle and close above the upper band and then return back again and closes inside the runway with a bearish candle, then a sell signal is generated. MACD indicator is used for justification.



Figure 16: Trend Reversal Strategy for Sell Signal

Strategy 3: M/W pattern strategy

W pattern (Buy Signal): In a downward trend, if the price makes a low peak outside the lower band and after bounce back inside and make another peak inside the runway then again the direction reverses and becoming from downward it become upward trend. MACD indicator is used for justification.

M pattern (Sell Signal): In an upward trend, if the price make a high peak outside the upper band and then bounce back inside and make another peak inside the runway, it is the indication of M-pattern and then the direction of trend becoming reversal (going from upward to downward) and this generates a sell signal. MACD indicator is used for justification



Figure 17: W pattern strategy for Buy signal



Figure 18: M pattern strategy for Sell signal

Strategy 4: Bollinger Squeeze strategy

Buy Signal: During low volatility in Bollinger band, if the macd indicator possess a sell signal, then positive breakout will occur which generates a buy signal.



Figure 19: Bollinger band Squeeze strategy for Buy Signal

Sell Signal: During low volatility in Bollinger band, if the macd indicator possess a buy signal, then negative breakout will occur which generates a sell signal.



Figure 20: Bollinger band Squeeze strategy for Sell Signal

V. RESULTS

We are testing this model on Reliance stock dataset. First we have to apply strategical analysis on the model. Then we have to apply ARIMA time series model for forecasting trend.

Below diagram shows the Bollinger band plot after computing Bollinger band attributes.



Figure 21: Output for Bollinger band indicator

In the above graph at the end point, the price candles are on the top of the 20-day moving average line. So, now we have to apply the strategies which are defined in the earlier section. Here as we see, after a downtrend the bearish candle closes below the lower band and the bullish candle closes below the lower band. So, we have to apply strategy 2, trend reversal strategy for buy signal. So, at the end point of the graph a buy signal is generates.

Below diagram shows the Moving Average Convergence Divergence diagram after computing MACD attributes.

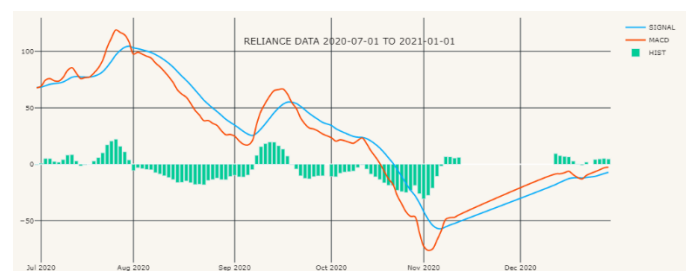


Figure 22: Output for Moving Average Convergence Divergence indicator

In the above graph, at the end after a short sell signal the macd line crosses above the signal line which generates a buy signal.

From the Technical analysis we get the result as Buy signal to continue. This is justified with forecasted trend using ARIMA time series model. Below diagram shows the forecasted trend using ARIMA model of Reliance stock dataset.

In the below figure, the red line indicates the forecasted Close values using ARIMA time series algorithm. At first the red line goes upwards means that it is making upward trend. This confirms that the generated signal is a Buy Signal.

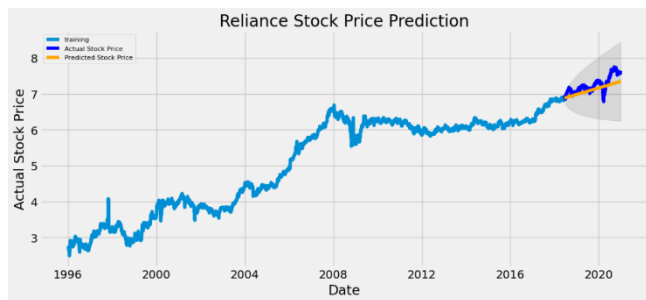


Figure 23: Output for Forecasting the Reliance stock dataset

Accuracy metrics for forecasted trend,

MSE: 0.03638599251868503
MAE: 0.1544972584912247
RMSE: 0.19075112717539844
MAPE: 0.02095856531328833

Around 2% of MAPE (Moving Absolute Percentage Error) implies the model is about 98% accurate in predicting the test set observations.

VI. CONCLUSION

In this study, an average individual may also buy or sell signals at the right time with gains and losses with a little basic knowledge. This is performed by integrating two main technical indicators, Moving Average Convergence Divergence and Bollinger Bands, to develop strategies, and then forecasting the trend with the ARIMA Time Series model. This approach is best suited for short-term trading. This model achieves an accuracy rate of 98% in forecasting the trend and MAPE of 2%.

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